

Planar cell polarity coordination in a cnidarian embryo provides clues to animal body axis evolution

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This analysis of the formation of the oral-aboral body axis in cnidarians, the sister group of bilaterians, is a significant and **fundamental** contribution to the field of Wnt signalling and planar cell polarity, particularly in our understanding of gradient formation, non-canonical Wnt signalling and Wnt-Frizzled interactions in cnidarians. The evidence supporting the conclusions is **compelling** and has the potential to contribute to a deeper understanding of the origin and evolution of Wnt signalling in cnidarians and metazoans in general. These findings, which are presented in a thoughtful and scholarly manner, will be of broad interest to developmental and evolutionary biologists.

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Abstract

Summary

Body axis specification is a crucial event in animal embryogenesis and was an essential evolutionary innovation for founding the animal kingdom. It involves two distinct components that coordinate to establish the spatial organisation of the embryo: initiation of cascades of regionalised gene expression and orientation of morphogenetic processes such as body elongation. Intense interest in the first component has revealed Wnt/ β -catenin signalling as ancestrally responsible for initiating regional gene expression, but the evolutionary origin of oriented morphogenesis has received little attention. Here, by addressing the cell and morphological basis of body axis development in embryos of the cnidarian *Clytia hemisphaerica*, we have uncovered a simple and likely ancestral coordination mechanism between Wnt/ β -catenin signalling and directed morphogenesis. We show that the ligand Wnt3, known to initiate oral gene expression via localised Wnt/ β -catenin pathway activation, also has a key β -catenin-independent role in globally orienting planar cell polarity (PCP) to direct morphogenesis along the oral-aboral axis. This PCP orientation occurs in two distinct steps: local orientation by Wnt3 and global propagation by conserved

core PCP protein interactions along the body axis. From these findings we propose novel scenarios for PCP-driven symmetry-breaking underlying the emergence of the animal body plan.

Introduction

The earliest animals are thought to have been multicellular assemblages such as the “choanoblastaea”, which evolved from a unicellular ancestor (Brunet and King, 2017; Maldonado, 2004; Nielsen, 2008; Seb  -Pedr  s et al., 2017). Subsequent evolutionary events that introduced a morphologically distinct primary body axis, i.e. breaking of tissue polarity symmetry, remain largely unaddressed (Fig. 1A). In the animal lineage, Wnt/ β -catenin-mediated specification of the primary body axis was likely an early innovation since it is found today across many species of both bilaterians (all of the animals’ groups showing bilateral symmetry) and cnidarians (their sister group, comprising corals, sea anemones, jellyfish etc) (Holstein et al., 2011; Loh et al., 2016; Petersen and Reddien, 2009). Current evidence is insufficient to assess whether this innovation predated the branching of the more distantly related ctenophore and sponge clades (J  ger et al., 2013; Reid et al., 2018). Importantly, however, transcriptional activation by β -catenin and its partner TCF at one pole cannot alone explain the morphological component of axial tissue patterning (Philipp et al., 2009). Rather, polarity information capable of directing axis development can be distributed globally across the whole embryo, as clearly illustrated by classic experimental manipulations in the hydrozoan *Clytia gregarium*, which led to the concept of global polarity (Freeman, 1981; Primus and Freeman, 2004).

Freeman’s global polarity concept retrospectively has much in common with planar cell polarity (PCP), the mechanism coordinating the cell polarity in the plane of epithelial tissues, later identified and unravelled genetically in *Drosophila* and vertebrates. PCP involves interactions between neighbouring cells, with each cell in a sheet, thus carrying polarity information. It is mediated by interactions of core PCP proteins: the Wnt receptor Frizzled (Fz), intracellular signal transducer Dishevelled (Dsh), transmembrane protein Strabismus/Van Gogh (Stbm), LIM-domain protein Prickle (Pk) and G-protein coupled receptor cadherin Flamingo (Fmi), to ensure tissue polarity and enable morphogenesis (Butler and Wallingford, 2017; Gray et al., 2011; May-Simera and Kelley, 2012; Seifert and Mlodzik, 2007; Vladar et al., 2009; Zallen, 2007). A conserved feature of PCP is the polarised localisation of the Stbm-Pk-Fmi complex on one side of the apical cortex of each cell and Fz-Dsh-Fmi complex on the other side. These two complexes interact between adjacent cells to coordinate PCP orientation of neighbouring cells. The PCP proteins are conserved across bilaterians and cnidarians, as well as in some sponge groups (Hale and Strutt, 2015; Lap  bie et al., 2011; Schenkelaars et al., 2016). Fz and Dsh are components both for core PCP and Wnt/ β -catenin signalling, so Wnt, Fz and Dsh are potentially key elements in coupling the pathways.

Two crucial issues for the developing embryo and the earliest animal ancestors are i) to orient PCP and thus morphogenesis consistently and globally across the whole embryo and ii) to coordinate PCP orientation with regionalised cell fate specification initiated by Wnt/ β -catenin signalling. These issues have not been addressed in the context of the primary body axis specification, while current examples of PCP orienting mechanisms from other tissues and animals show considerable variability (Butler and Wallingford, 2017).

We address here the coupling of Wnt signalling and PCP during body axis specification in a cnidarian in order to shed light on the roles of PCP in the evolutionary emergence of the metazoan body axis, exploiting the jellyfish model species *Clytia hemisphaerica* (Houlston et al., 2022). Previously, we showed that maternal mRNAs for the ligand CheWnt3 (orthologue of bilaterian Wnt3, hereafter called Wnt3) and its receptors are localised with respect to the animal-vegetal polarity of the unfertilised *Clytia* egg (Fig. 1B), and activate Wnt/ β -catenin signalling in a future

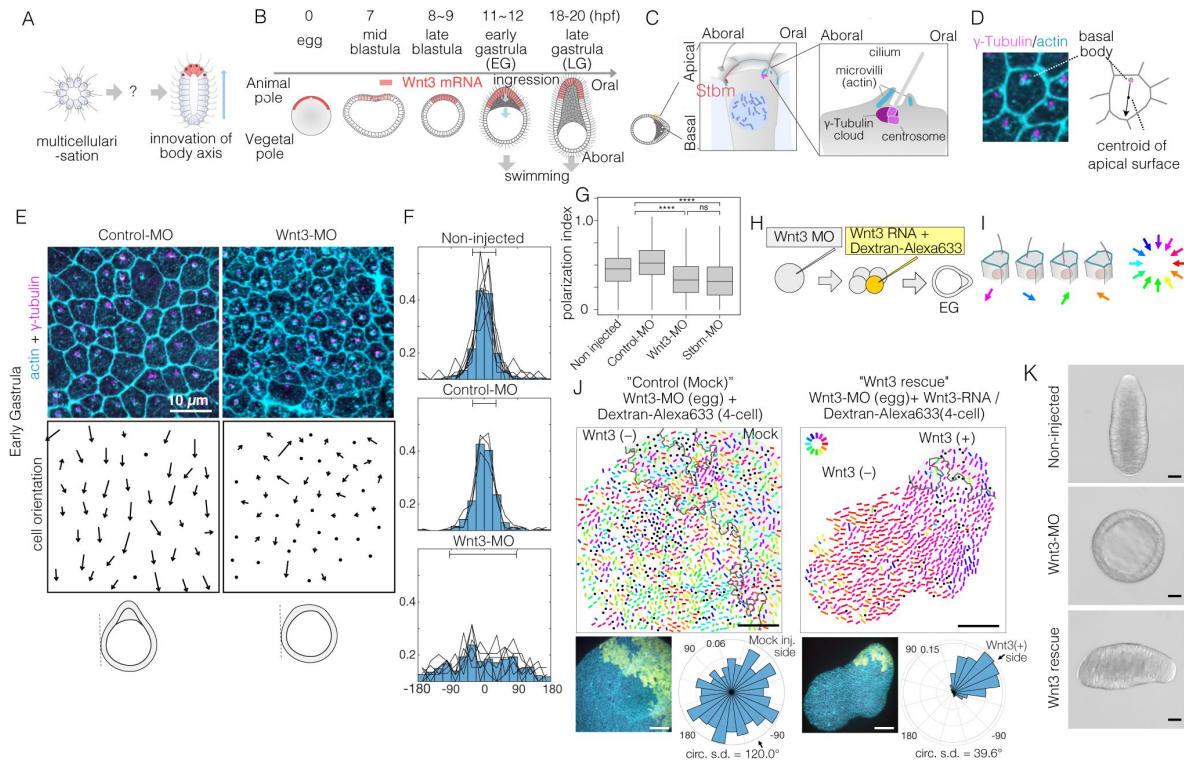


Figure 1

Local Wnt3 globally orients the body axis tissue polarity and morphogenesis

(A) A key unanswered question of animal evolution is the origin of body axis patterning mechanisms, allowing initial multicellular assemblages to position specialised cell types (red) and coordinate cell polarity (represented in this diagram by oriented cilia and elongated shape) (B) Key stages of the early morphogenesis of *Clytia* embryos in relation to Wnt3 mRNA localisation (red). At the early gastrula (EG) stage, ectodermal cilia have formed, PCP has developed along the future oral-aboral axis, and the pointed morphology of the animal pole morphology can first be distinguished. Most axial elongation takes place during gastrulation. See also Supplementary Figure 1. (C) Schematic representation of planar cell polarity in a single epidermal cell at the EG stage. Basal bodies are positioned on the oral side at the apical cortex (translational polarity). (D) Confocal image of the apical surface of a single epidermal cell stained for actin (cyan, phalloidin labelling) and γ -tubulin immunostaining (magenta), and a representation of the PCP vector, defined by the basal body position and its mirror image across the apical surface centroid as the initial and terminal points, respectively. (E) Local PCP coordination in Control-MO and Wnt3-MO-injected embryos at the EG stage. bar= 10 μ m. Top) Confocal images as in D. Bottom) PCP vector representation of the cells in the images. Dots represent cells with no clear polarisation or mitotic cells. Apical cell contours are wavy in Wnt3-MO embryos. (F) Distribution of cell orientation in individual scans (lines) and their average (histogram) for non-injected (373 cells, 7 images), control-MO injected (304 cells from 3 images) and Wnt3-MO-injected (225 cells from 6 images). The average orientation is defined as the origin (0°) for each scan. Horizontal bar: standard deviation. (G) The PCP polarisation index (degree of the polarisation of cells) by the length of the PCP vector, normalised by the cell size. See supplementary method for the definition (**** $p < 0.0001$, Mann-Whitney U test). (H) Experimental procedures of the Wnt3-rescue experiment. Wnt3MO is injected into unfertilised eggs, then rescue mRNA solution with fluorescently labelled 10 kDa dextran is injected into a 4-cell stage blastomere. (I) Circular colour-code representation of the PCP. (J) Wide range PCP coordination observed in Wnt3-rescue experiments. Left: Mock rescue control by water injection (N=9). Right: Rescue by Wnt3 mRNA injection (N=27). Each set consists of three illustrations. top: colour-wheel representation of the PCP. Cells derived from the Wnt3 or water-injected blastomere are at the top-right of each image, indicated with grey shading. bar= 50 μ m. bottom-left: confocal thumbnail image. cyan: actin; yellow: Dextran-Alexa647(injected blastomere derived cells). bottom-right: radar plot of the cell orientation. The arrows in the radar show the circular mean of the PCP orientation within the single scan. circ s.d.: circular standard deviation. (K) Axial morphology of normal, Wnt3-MO injected and Wnt3 rescued embryos at late gastrula (LG) stage. bar=50 μ m. Elongated body axis morphology was restored so that cells with injected Wnt3 mRNA were located at the induced oral (posterior). See also Fig. 3C.

oral (posterior) domain of the developing embryo to control gene expression along the developing primary body axis (oral-aboral: OA) (Momose et al., 2008 [↗](#); Momose and Houliston, 2007 [↗](#)). In parallel, conserved core PCP proteins, including Stbm orthologue CheStbm (Stbm), are responsible for establishing tissue polarity (PCP) in the epidermis to support directed larval swimming via cilia beating along the oral-aboral axis, and are also required for elongation along the primary body axis during gastrulation (Momose et al., 2012 [↗](#)). Stbm protein is localised on the aboral side of each cell (Momose et al., 2012 [↗](#)) as expected from *Drosophila* and vertebrate PCP deployment mechanisms.

In the current work, we explore the hypothesis that Wnt3, as well as activating Wnt/ β -catenin signalling (Momose et al., 2008 [↗](#)), provides the initial cue to direct global orientation of PCP and thereby define the *Clytia* embryonic body axis both molecularly and morphologically. A first hint of this from our previous work was that the characteristic and complete absence of embryonic axis elongation observed following Wnt3 knockdown by Morpholino (Wnt3-MO) injection could be restored by injection of synthetic Wnt3 mRNA into single blastomeres during early development (Momose et al., 2008 [↗](#)). Importantly, Wnt3 is the sole Wnt mRNA detected before the early gastrula (EG) stage, the period when axial morphology and epidermal PCP are established, with other Wnt genes being expressed downstream of Wnt3 or at subsequent stages of larval development (Lapébie et al., 2014 [↗](#); Momose et al., 2008 [↗](#)) (Fig.1B [↗](#) Fig. S1). This greatly facilitates the assessment of Wnt ligand function in directing PCP, in contrast to traditional model species where multiple Wnt ligands have parallel roles during embryogenesis. The studies reported here reveal how local Wnt3 cues orient the global axial PCP and PCP-driven oriented morphogenesis in the *Clytia* embryo. Two discrete steps define the PCP orientation: a local cueing step with orally localised Wnt3 signal and a global propagation and coordination step mediated by core PCP proteins. Interestingly, while the role of Wnt3 is the instructive cue for PCP orientation and thus morphological axis development during *Clytia* normal development, other cues, including mechanical strains, can orient PCP. A similar variety of PCP orientation cues has been reported in *Drosophila* and vertebrates. Our findings allow us to propose a ‘PCP-first’ model for body axis symmetry-breaking in the metazoan ancestors, in which the self-organising activity of the PCP pathway allowed body axis innovation in early animal ancestors.

Results

A local Wnt3 cue orients global PCP

We first analysed in detail PCP in Wnt3-depleted embryos at the EG stage, the earliest stage that shows morphological OA polarity (Fig.1B [↗](#), S1). PCP manifests as structural and positional asymmetries of the ciliary basal bodies, which are consistently positioned towards the oral side of the apical cortex. As previously (Momose et al., 2012 [↗](#)), we quantified epidermal PCP orientation by assessing the basal body positioning (Fig.1D [↗](#)). In unmanipulated or control-MO-injected embryos, cells showed coordinated PCP, with 85% of cells orienting within $\pm 45^\circ$ (s.d. 33 and 34° respectively) of the average orientation. In embryos injected with Morpholino antisense oligonucleotides (Wnt3-MO) to block Wnt3 translation, PCP was oriented nearly completely randomly (Fig.1E, F [↗](#)) (s.d. 84°, comparable to 95° for Stbm-MO embryos (Momose et al., 2012 [↗](#))), and individual cells were less polarised (Fig.1G [↗](#)), reflecting the complete lack of axial morphology. The apical cell boundary was wrinkled in Wnt3-MO-injected cells (Fig. 1E [↗](#)), as observed following Fz1-MO, Dsh-MO or Stbm-MO-injections (Momose et al., 2012 [↗](#)), indicative of markedly reduced cortical tension. These analyses demonstrate that Wnt3 depletion causes PCP defects equivalent to the knockdown of the core PCP proteins (Momose et al., 2012 [↗](#)). This supports the hypothesis that Wnt3 provides a local cue to orient global PCP, which is responsible for the development of cilia polarity and axially elongated morphogenesis in *Clytia*.

To test this hypothesis, we injected 100 ng/μl Wnt3 mRNA and fluorescently labelled Dextran into one blastomere at the four-cell stage in Wnt3-MO embryos, which we call the “Wnt3-rescue” experiment (**Fig. 1H, J**). At 10 ng/μl Wnt3 mRNA, axis restoration was random and either quasi-complete axis restoration and no axis. At the EG stage, we quantified PCP in a broad area (246 x 246 μm) and represented it by bars with a circular colour code (**Fig. 1I**). Control mock injections did not affect the PCP defects caused by Wnt3-MO. In contrast, coordinated PCP was restored by Wnt3 rescue in a non-autonomous manner (**Fig. 1J**). Restored ciliary alignment was observed at least 200 μm away from the Wnt3 mRNA-positive area. The Wnt3 source likely influences PCP at even further distances as a virtually complete morphological axis was restored by the late gastrula stage, with the Wnt3-mRNA lineage forming a new oral pole (**Fig. 1K**) as shown previously (Momose et al., 2008). Locally expressed Wnt3 signal is thus necessary and sufficient to orient the PCP along the whole body axis.

Rapid global PCP development precedes basal body alignment

To understand the initial process by which PCP becomes coordinated by Wnt3, we examined stages preceding EG, again using basal body localisation as the read-out. We reasoned that if PCP propagation occurs through successive cell interactions, cell alignment might spread as a wave across the embryo. PCP was detectable from the mid-blastula stage (8-9 hpf **Fig. 2A-D**), slightly after the onset of the zygotic Wnt3 expression (Momose et al., 2008) and epithelialisation of the blastula cells (Kraus et al., 2020). PCP was slightly better coordinated in the region proximal to the Wnt3-positive area (**Fig. 2C, D**). We did not, however, detect any wavefront of local cell alignment; rather, the PCP gradually increased in coherence across the embryo over time (**Fig. 2A, B**). This was confirmed by live imaging, injecting synthetic mRNAs in the egg to visualise cell contours with PH-Venus and basal bodies with Poc1-mCherry (**Fig. 2E**, **Movie S1**). The movies further revealed active mitosis in the blastodermal cells. A weak bias in cleavage orientation was observed along the OA axis (**Fig. 2F**). During mitosis, duplicated centrioles from the ciliary basal bodies are deployed in the mitotic spindles (**Fig. 2E**, **Movie S1**) such that basal body positioning becomes scrambled immediately after mitosis. Subsequently, the displaced basal bodies migrate unidirectionally toward the oral side of each cell, where the cilium is re-established, a clear indication that PCP was already in place across the tissue (**Fig. 2E, G, H**). Together these observations indicate that PCP is established at the mid-blastula stage, presumably via canonical localisation of core PCP proteins to opposite cortices of each cell through continuous Fz-Stbm interaction. The protein partitioning then manifests as asymmetric basal body positioning. Localised PCP proteins may attract basal bodies to the oral cortex of the cell as described in ascidian embryos (Negishi et al., 2016). During subsequent stages, PCP is either maintained (Bellaiche et al., 2004) in each cell during mitosis or rapidly restored through interactions with neighbouring cells (Devenport et al., 2011).

Wnt3 orients PCP in a β -catenin-independent manner

We previously showed that local Wnt3 expression is necessary and sufficient to activate the Wnt/ β -catenin pathway in the oral pole of the *Clytia* embryo (Momose et al., 2008). We next examined whether the action of Wnt3 for PCP orientation and morphogenesis is mediated by the Wnt/ β -catenin pathway. First, we showed that PCP orientation by Wnt3 is independent of β -catenin: ectopic Wnt3 could orient PCP in the Wnt3-rescue experiment even in the presence of a dominant-negative form of CheTCF (dnTCF) (**Fig. 3A**), which blocks downstream transcription of the β -catenin pathway. Gastrulation was completely prevented, confirming that the Wnt/ β -catenin pathway was inactive (**Fig. 3C, D**, **Fig. S2**). Furthermore, body axis elongation was partially restored at the early gastrula stage in the absence of an active β -catenin pathway (**Fig. 3C**), although they did not develop pointed morphology of the oral pole nor elongate further (**Fig. 3D**, **Fig. S2**). These results support a model in which axial elongation requires both PCP coordination and gastrulation (migration and intercalation of ingressed endodermal cells), as suggested previously (Kraus et al., 2020; Sande et al., 2020). Conversely, we found that

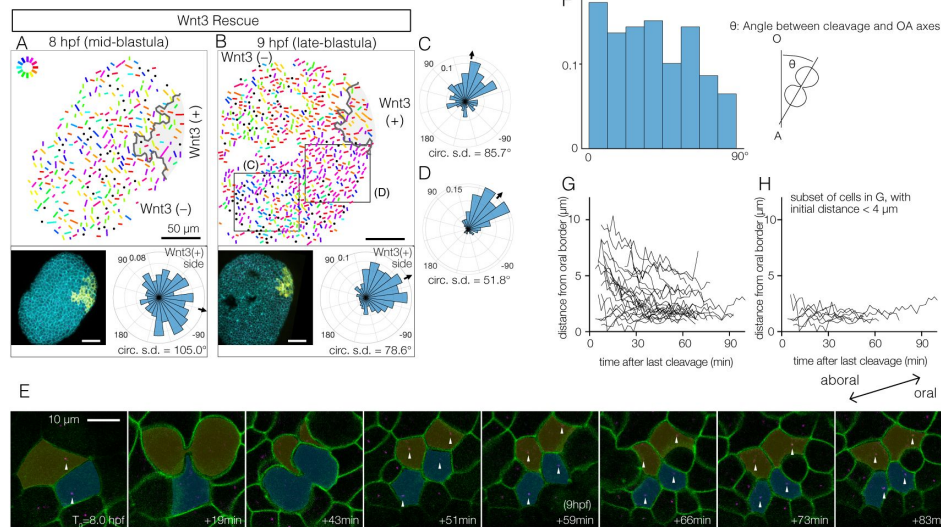


Figure 2

Wnt3-driven PCP orientation precedes visible cell alignment.

(A, B) PCP orientation in Wnt3 rescue embryo at the mid-blastula stage (A: 8 hpf, N=10) and late-blastula stage (B: 9 hpf, N=10). Graphical PCP representations and radar histogram plots are as in [Fig. 1](#). (C, D) Radar plot of PCP in distal (C) and proximal (D) areas of the region shown in (B). (E) Basal body displacements after cell divisions. Bar=10 μ m. Cell membranes labelled by PH-Venus (green) and basal bodies with Poc1-mCherry (magenta). Arrows indicate basal bodies positioned on opposite sides of cleavage planes located after being engaged to the spindles. Aborally displaced basal bodies migrate toward the oral side. The average PCP defined the oral-aboral axis. (F) Distribution of the angles between the cleavage orientation axis to the OA axis defined by the PCP orientation (0-90°, n=131). Cleavages along the OA axis (0-45°) are favoured (* p<0.05 chi-square test). (G) Directional migration of the basal body toward the oral cell boundary after the cell cleavage. Each line indicates the distance of a single basal body to the oral edge of the cell after the last cleavage. Basal bodies migrate toward the oral cell boundary, suggesting PCP has been established by the time PCP becomes structurally manifest. (H) Subset of (G) where the initial distance to the oral edge is less than four micrometres after the cell division.

constitutively active β -catenin (CA- β -cat) mRNA was not able to restore PCP coordination nor elongation of Wnt3-depleted embryos (**Fig.3B**) but did restore β -catenin-dependent gastrulation (Fig. S2). Taken together, these experiments demonstrate that global PCP orientation along the *Clytia* OA axis by Wnt3 is not mediated by the Wnt/ β -catenin pathway. Wnt3 thus has dual independent roles: PCP orientation and Wnt/ β -catenin activation.

Global PCP is established in two distinct steps

We next tested the contribution of the core PCP proteins CheStbm and CheFmi (hereby called Stbm and Fmi) to linking PCP orientation to the local Wnt3 source. As expected, locally confined expression of Wnt3 mRNA did not restore global PCP coordination in Stbm-MO or Fmi-MO embryos (**Fig.3E, F**). Importantly, however, polarity was restored in the area immediately adjacent to the Wnt3 mRNA-positive cell patch. PCP orientation seemed less coordinated within the Wnt3-mRNA-positive area. The effect covered a distance of a few cells, potentially derived from cells directly in touch with the Wnt source at earlier stages. This observation raised the possibility that global PCP orientation in *Clytia* embryos is established in two steps: Stbm/Fmi-independent PCP orientation by local gradient or contrast of Wnt3 activity at the oral end, followed by global PCP propagation through the intercellular interaction of core PCP proteins (two-step model), as predicted by the domineering non-autonomy property of mosaic PCP mutations (Amonlirdviman et al., 2005).

Alternatively, Wnt3 may diffuse from the oral Wnt3-expressing cells (**Fig.1A**) to form a long-range gradient along the OA-axis and orient the PCP (Wnt-gradient model). Wnt activity gradients have been shown to control or orient PCP in mice (Gao et al., 2011; Minegishi et al., 2017).

To distinguish these two models, we created mosaic embryos with a small patch of core-PCP negative cells within Wnt3-rescued embryos.

To achieve this, we transplanted single 32- or 64-cell stage blastomeres from Stbm-MO/Wnt3-MO or Fmi-MO/Wnt3-MO injected embryos to Wnt3-rescued embryos with Dextran lineage markers (**Fig. 4A-E**). The two-step model predicts that the PCP orientation would make a detour around the Stbm-MO/Fmi-MO mosaic area; in contrast, under the Wnt-gradient model, PCP would be affected only in the mosaic area. We found that in the Stbm-MO or Fmi-MO mosaic Wnt3-rescued embryos, PCP was correctly oriented or slightly disturbed within the Stbm-MO patch while often depolarised in the Fmi-MO patch. Outside the Stbm-MO or Fmi-MO patches, PCP orientation was inverted on the aboral side of the patches unless these were in close proximity to the Wnt3-mRNA patches. Stbm- or Fmi-defected cells thus exhibited domineering non-autonomy effects. This is consistent with the intercellular Fz-Fmi and Sbm-Fmi interaction model shown in *Drosophila* and vertebrates and the localisation of Stbm protein on the aboral side of each epidermal cell in *Clytia* (Momose et al., 2012). PCP was oriented normally in the cells on the oral or lateral sides of the patches (**Fig. 4B, D**). PCP orientation from oral to aboral detouring around the Stbm-negative patches was similarly observed when mosaics were generated by injection of Stbm-MO into a single blastomere in a Wnt3-rescue embryo at the 16-cell stage (**Fig.4F, G**). These observations strongly support the two-step model in which Wnt3 acts locally to orient PCP, and PCP coordination is then transmitted from oral (i.e. Wnt3-positive lineage) to aboral through core PCP interactions. They are not compatible with the alternative model that a long-range Wnt3-gradient orients PCP.

Mechanical cues can orient PCP

Transplantation of Wnt3-MO blastomeres into Wnt3-rescued host embryos at the 32- or 64-cell stage did not cause any PCP defects in most cases. However, in a few cases (N=3 out of 8), reorientation of PCP toward the transplanted donor lineage was observed (**Fig.5A-E**). In these cases, the donor cells were incompletely incorporated into the host ectodermal epithelium (**Fig.**

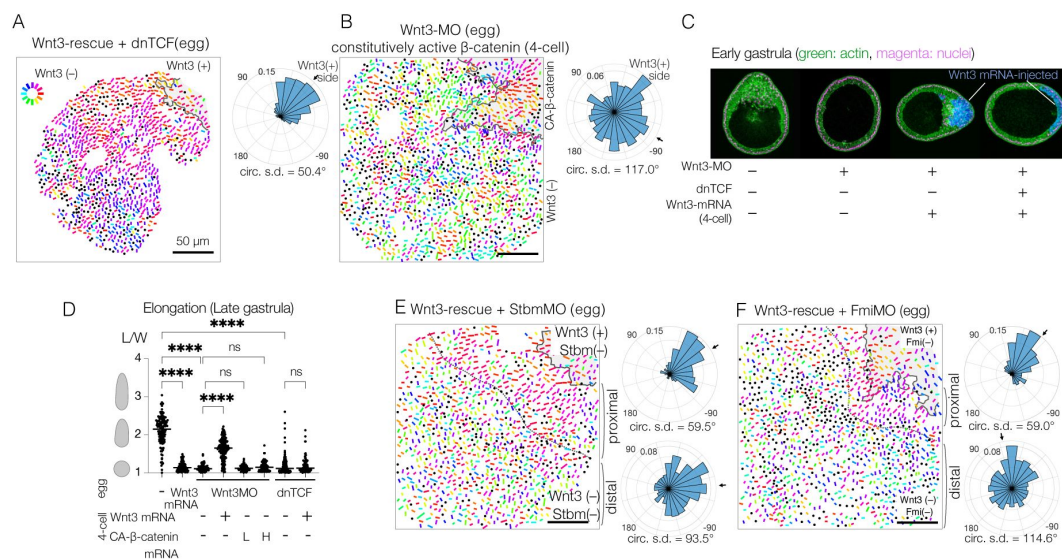


Figure 3

PCP orientation by Wnt3 is Wnt/β-catenin-independent, and its propagation across the body axis requires Stbm and Fmi.

(A) Wnt3-rescue with an additional injection of a dominant-negative form of TCF (dnTCF) into the egg, N=10. (B) Wnt3-rescue by the constitutively active form of β-catenin (CA-β-cat) instead of Wnt3 mRNA, N=7. (C) Onset of axial morphology in EG embryos. Localised Wnt3 is necessary and sufficient for the onset of the elongation toward its source. Green: actin (phalloidin), magenta: nuclei (To-Pro3) and blue: Wnt3-mRNA/Dextran injected cell lineage. (D) Roles of Wnt3 and β-catenin in the axial elongation in LG embryos. The elongation was measured by the elongation index (L/W): the primary axis length (L) divided by its perpendicular length (W), **** $p < 0.0001$, Mann-Whitney U test). Both local Wnt3 expression and β-catenin/TCF-dependent gastrulation were required for axial elongation. See supplementary Figure 3. (E, F) Wnt3-rescue experiments with additional injections of (E) Stbm-MO (N=23) and (F) Fmi-MO (N=6). Radar plots of the PCP orientation show the manually defined proximal area (Proximal: between the dotted line and the Wnt3-positive region) and distal area (Distal: opposite area to the Wnt3-positive region across the dotted line). PCP is correctly polarised without Stbm or Fmi close to the Wnt3-injected cell clone but not in the distant area.

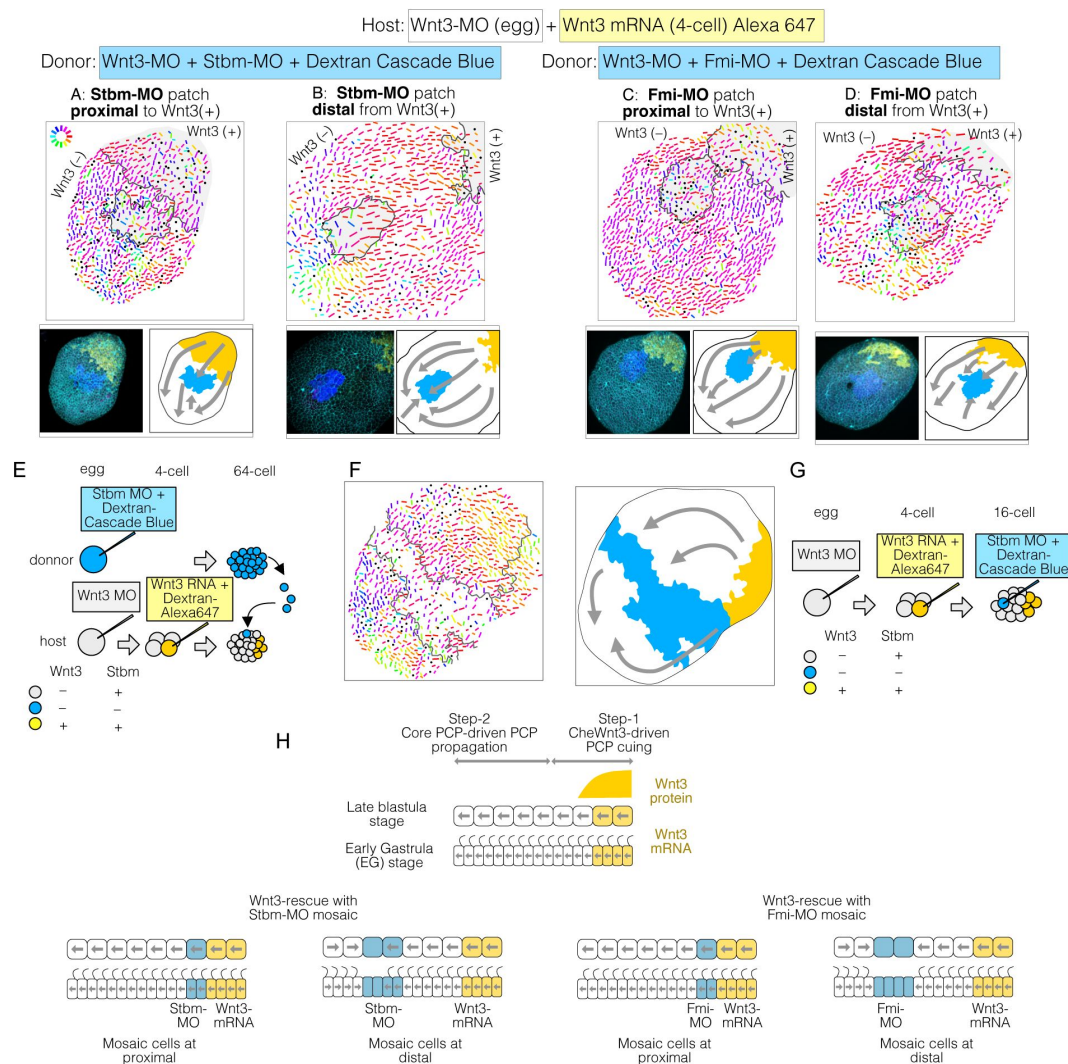


Figure 4.

PCP behaviour in Stbm/Fmi-mosaic/ Wnt3-rescue experiments supports the two-step model for global PCP orientation

(A-D) Wnt3-rescued early gastrula embryos, including Stbm(-) or Fmi(-) mosaic patches made by blastomere transplantation. (A, B) blastomeres from Wnt3-MO/Stbm-MO double-injected embryos were transplanted as the donor into Wnt3-rescued host embryos at the 64-cell stage. They happened to be incorporated in close/proximal (A: 658 cells, N=4) or far/distal (B: 757 cells, N=4) positions with respect to the Wnt3 source, respectively. (C, D) The same experiment was conducted with Wnt3-MO/Fmi-MO-injected blastomeres as the donor, incorporated in proximal (C: 984 cells, N=3) and distal (D: 649 cells, N=3) positions, respectively. (E) Schematic drawing of the experimental procedure of the transplantation. (F) Wnt3-rescued early gastrula embryos with Stbm(-) mosaic patch generated by additional injection of Stbm-MO at the 16-cell stage (N=1). (G) Schematic drawing of the experimental procedures for (F). The flow of the PCP orientation is indicated by grey arrows in the bottom right drawings in (A-D) and right in (G). (H) Graphical summary of Stbm-MO and Fmi-MO mosaic experiments. The graphical representation and thumbnail confocal images indicate Wnt3-positive lineages and Stbm-MO or Fmi-MO-injected lineages in yellow and blue. All vector representations are as in [Figure 1](#).

5D [↗](#)), such that the donor cells remained as a protrusion with a “rosette” (radial) arrangement. In these embryos, host and donor-derived epithelial cells acquired an elongated morphology around the circumference of the smooth host-donor boundary (**Fig.5B** [↗](#)). Although the grafted tissue influenced the orientation of PCP in the surrounding host cells, the direction was opposite to that driven by Wnt3-expressing sources. The rosette-like transplants thus operate as artificial “aboral” cues, in contrast with Wnt3 sources, which provide an oral cue (**Fig. 5C** [↗](#)). Unlike Stbm-MO or Fmi-MO mosaic embryos, cell polarity was continuous across the host-transplant border in these Wnt3-MO to Wnt3-MO transplantations (**Fig.5A** [↗](#)), suggesting the core PCP protein interaction was not disturbed at the boundary and thus not likely due to the nonautonomous effect of PCP defects. Epithelial ‘scarring’ can thus act to orient PCP, perhaps via discontinuities in cell adhesion or mechanical strain. PCP re-orientation by mechanical strain has been reported in wing PCP repatterning in *Drosophila* and for the orientation of *Xenopus* epidermal motile cilia (Aigouy et al., 2010 [↗](#); Chien et al., 2022 [↗](#); Guirao et al., 2010 [↗](#); Mitchell et al., 2007 [↗](#); Olguin et al., 2011 [↗](#)). We also performed transplantations into Wnt3-MO-only hosts but did not obtain any such rosettes, so we could not determine whether Wnt has a permissive role for PCP in this context. Nevertheless, we can conclude that any local PCP asymmetry can potentially act as a PCP orientation cue via core-PCP proteins, and thus again is consistent with the two-step orientation model.

To explore further whether mechanical strain can orient PCP, we disrupted the morphology of the embryo by expressing dominant negative and constitutively active forms of *Clytia* RhoABC, the single cnidarian orthologue of the vertebrate Rho GTPases including RhoA. Late gastrula embryos expressing dominant negative RhoABC showed defects in axial elongation, consistent with a role in morphogenesis. Conversely, embryos injected with constitutively active RhoABC showed a scrambled axial morphology with ectopic protrusions induced along the major axis (**Fig.5F** [↗](#)). The induced protrusions appeared prior to gastrulation. PCP orientation was locally coordinated along the micro-axes of each protrusion so that they became extra aboral poles with respect to PCP (**Fig.5G, H** [↗](#)), while PCP orientation along the major body axis was globally maintained. Enhanced cortical actin fibres were arranged around the circumference of the induced protrusion (**Fig. 5I** [↗](#)). This method to provoke mechanical deformation of the embryo thus generated protrusions that mimicked the morphological extension and circumferential cell arrangement seen in the transplantation-induced rosettes, with in both cases PCP orientation aligning along the experimentally formed ectopic mini-axes.

Discussion

In this study, we have shown that during embryogenesis in the cnidarian *Clytia*, the ligand Wnt3, known to initiate oral gene expression via Wnt/ β -catenin pathway activation, also provides β -catenin-independent cues to orient PCP globally in the embryo and thereby direct morphogenesis along the oral-aboral axis. PCP orientation occurs in two distinct steps: local orientation by Wnt3 and global propagation via conserved core PCP machinery. These findings demonstrate deep conservation of PCP orientation and tissue morphogenesis mechanisms in eumetazoans, and allow us to propose novel scenarios for PCP-driven axis symmetry-breaking underlying the emergence of the animal body plan.

A single Wnt ligand coordinates molecular and morphological axis determination

Wnt/ β -catenin signalling (Holstein et al., 2011 [↗](#); Loh et al., 2016 [↗](#); Petersen and Reddien, 2009 [↗](#)) and PCP pathway components (Gray et al., 2011 [↗](#); Kumburegama et al., 2011 [↗](#); Momose et al., 2012 [↗](#); Wallingford, 2012 [↗](#); Zallen, 2007 [↗](#)) are involved respectively in embryonic body axis specification and morphogenesis across many metazoan species, but how these aspects are linked has been complicated to address. For instance, in vertebrates, Wnt5a and Wnt11 play crucial roles during body axis morphogenesis, (Heisenberg et al., 2000 [↗](#); Kilian et al., 2003 [↗](#); Tada and Smith,

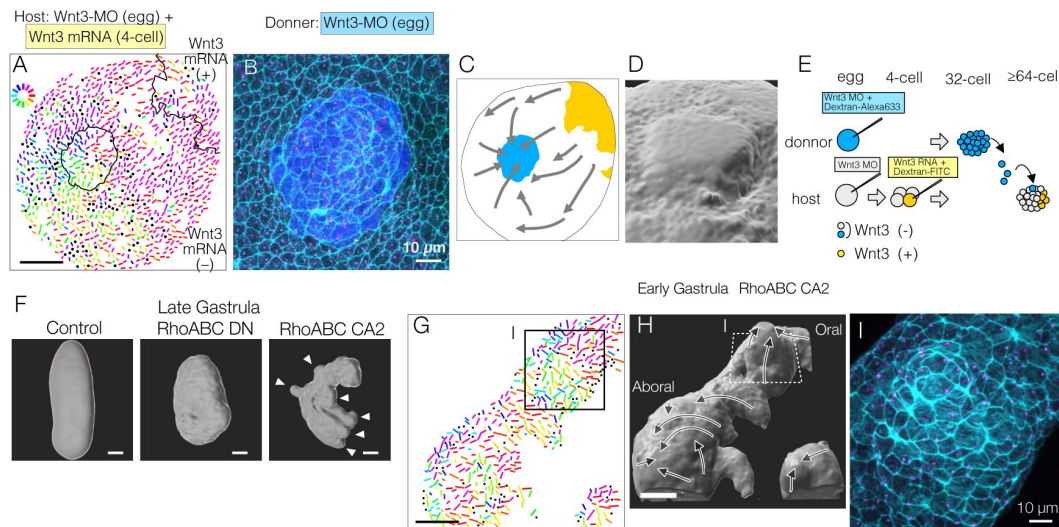


Figure 5.

Mechanical strains can reorient PCP.

(A-E) PCP orientation induced by transplantation from a Wnt3-MO donor into the Wnt3-MO background of a Wnt3-rescue host. (A) Incomplete incorporation of host and donor cells into a smooth epidermis (N=3/8) causing long-range PCP orientation, as if the donor acts as an aboral cue, in addition to the Wnt3 oral cue. (A) PCP representation by colour-coded bars. See also the legends for [Figure 1](#) (B) Confocal image of the apical cortex of the epidermis showing the rosette structure caused by incomplete incorporation of the donor cells. Both host and donor cells are elongated around the graft boundary. (C) Graphical summary of the PCP orientation: Wnt3-mRNA lineage in Wnt3-rescue host in yellow, Wnt3-MO graft in blue. (D) 3D reconstruction of the rosette structure from the confocal images by contouring cortical actin signals. (E) Schematic representation of the transplant experiment. (F) 3D representation of the late gastrula morphologies induced by mRNA injection of dominant negative form (RhoABC-DN) and constitutively active forms (RhoABC-CA2) of the PCP effector RhoABC, which caused reduced elongation and extra protrusions, respectively. The overall oral-aboral axis remained distinguishable based on morphology and gastrulation. (G-I) PCP is coordinated consistently with respect to the induced protrusions caused by RhoABC-CA2. (G) Colour-wheel representation of PCP in the RhoABC-CA2 injected early gastrula embryos; the presumed oral pole is to the top-right. (H) 3D reconstruction of the tissue morphology for the same specimen, with PCP flow represented by arrows. (I) Confocal image of the apical cortex of the epidermis at the induced protrusion, corresponding rectangles in G and H. Cortical actin fibres are circumferentially organised around the protrusion. Bars are 50 μm except for 10 μm in B and I. See [Figure 1](#) for the circular colour-code representation of PCP orientation. cyan: actin; magenta: γ -tubulin; blue: donor lineage marker Dextran for B and I

2000 [↗](#); Wallingford and Harland, 2001 [↗](#)) but it is not straightforward to demonstrate whether they provide positional cues to orient PCP along the body axis, while they do in other contexts (eg. orientation of ciliated node cells and muscle fibre) (Gros et al., 2009 [↗](#); Minegishi et al., 2017 [↗](#)). During *Clytia* embryogenesis, a single Wnt ligand directs both molecular and morphological aspects of axis formation. Wnt3 translated from maternally localised mRNA controls axial gene expression along the body axis via β -catenin regulation (Momose et al., 2008 [↗](#)). This dual role of the Wnt3 elegantly ensures the coherence of gene expression along the developing OA axis with morphogenetic aspects such as embryo elongation during gastrulation and beating of ectodermal cilia for directed locomotion. Maternal XWnt11 in *Xenopus* activates the Wnt/ β -catenin pathway (Tao et al., 2005 [↗](#)), and it is also implicated in the β -catenin-independent role to control convergent extension (Tada and Smith, 2000 [↗](#)). The dual roles of a Wnt ligand for PCP and Wnt/ β -catenin pathway may be a common strategy in vertebrates and cnidarians.

Two-step PCP orientation, initiated by CheWnt3 and propagated by core PCP interactions

The experiments presented here reveal two distinct steps of Wnt3-driven PCP and axial morphology in *Clytia* embryos (**Fig.6A** [↗](#)): A Wnt3-dependent local PCP orientation (cueing) step is followed by a propagation step mediated by core PCP proteins. This resembles the “domineering non-autonomy” phenomenon described in *Drosophila* mosaic mutants for core PCP genes (Amonlirdviman et al., 2005 [↗](#)). How Wnt3 locally orients PCP in the cueing step in *Clytia* embryos remains to be understood. Locally restricted PCP orientation by Wnt3 under the presence of Stbm-MO or Fmi-MO suggests a local gradient, or contrast, of Wnt3 acts as the orientation cue. Indeed, axis polarity is lost under uniform expression of Wnt3 (Momose et al., 2008 [↗](#)). A likely candidate mediating Wnt3 cue is CheFz1 (a single orthologue to vertebrate Fzd1/2/7/3/6), whose predicted localisation is on the oral side of each cell, opposite to the aboral accumulation of Stbm. The role of CheFz1 for mediating PCP orientation remains to be revealed, despite the effect of CheFz1 knockdown in PCP coordination (Momose et al., 2012 [↗](#)). Other Wnt3 receptor candidates are CheFz2 (Fzd5/Fzd8) and CheFz4 (Fzd4). An attractive hypothesis is that the local Wnt3 gradient, or contrast, anchors one of the Frizzled receptors at the oral side of the cells adjacent to the Wnt3-positive area in a mechanism similar to zebrafish Wnt11 that locally accumulates Fz7 in the plasma membrane (Witzel et al., 2006 [↗](#)). A further implication of the two-step model and the domineering non-autonomy character of PCP proteins is that PCP orientation may act as a morphogen-independent body axis maintenance mechanism in certain regeneration contexts in cnidarians (Livshits et al., 2017 [↗](#); Sinigaglia et al., 2020 [↗](#)). Established local PCP coordination may be propagated to restore the body axis in regeneration. In this context, it would be interesting to investigate the mechanism by which core PCP modulates axial gene expression. In *Clytia*, PCP defects resulting from Stbm knockdown led to changes in gene expression at the early gastrula stage, notably affecting cells in the oral pole that ingress during gastrulation (Lap  bie et al., 2014 [↗](#)). Our work also connects to historical hypotheses regarding hydrozoan embryonic axis patterning mechanisms. Nearly a hundred years ago, Georges Teissier proposed that egg polarity is either maintained or re-established as a non-localized polarity within the embryo, thereby defining the body axis (Teissier, 1931 [↗](#)). Later, Gary Freeman revisited Teissier’s experiments and suggested that this polarity, somewhat consistent with the modern concept of planar cell polarity, is directly inherited from egg polarity (Primus and Freeman, 2004 [↗](#)). Freeman considered this an alternative hypothesis to the maternal axis determinant model, which was later demonstrated in *Clytia* and other hydrozoans (Momose et al., 2008 [↗](#); Momose and Houliston, 2007 [↗](#)). Our work highlights PCP propagation as a mechanistic explanation of Teissier and Freeman’s non-localised polarity at gastrula and planula stages, while the two-step axis patterning model provides a link to localised Wnt/ β -catenin pathway activation triggered by maternal determinants.

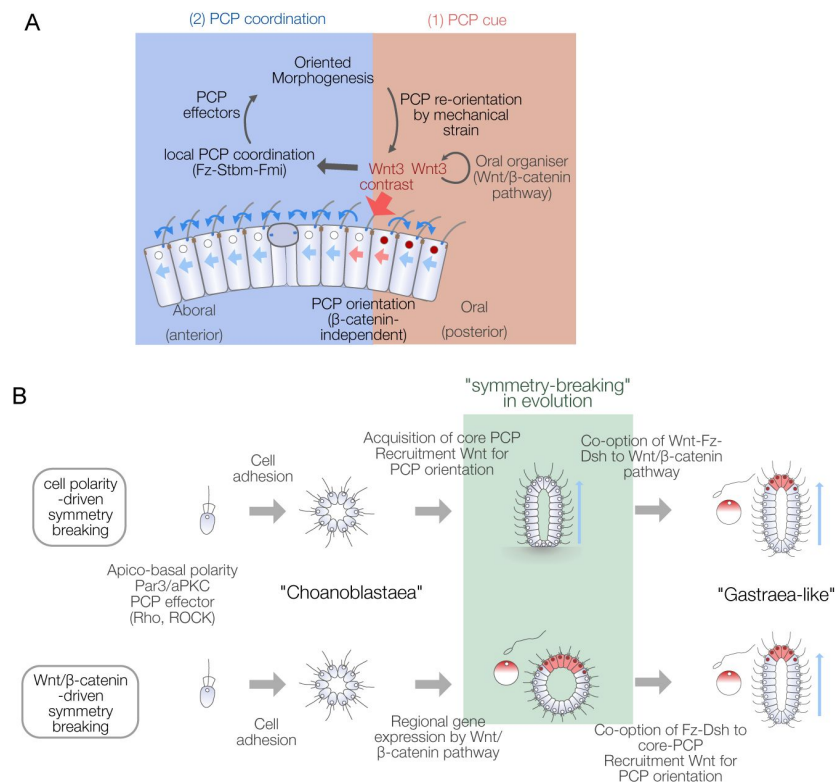


Figure 6.

Models for body axis symmetry-breaking during *Clytia* embryogenesis and PCP-driven axis innovation scenarios in metazoan evolution

(A) Two-step PCP axis orientation by local Wnt3. Local expression of Wnt3 protein at the oral end of the *Clytia* embryo orient epidermal cell polarity near the Wnt3-positive area. This process does not require Wnt/β-catenin signalling. Local PCP orientation propagates to the aboral side through the action of core PCP proteins between neighbouring cells. Wnt3 expression is likely maintained by Wnt3/β-catenin feedback regulation. (B) Two possible metazoan body axis evolution scenarios from a choanoblastaea, a multicellular and non-polarised metazoan common ancestor. In the "Wnt/β-catenin-first" scenario (bottom), the choanoblastaea first acquires a mechanism to activate the Wnt/β-catenin pathway locally. Stably locally maintained Wnt ligand expression is then recruited by preexisting but non-directed core Fz/PCP interactions to orient PCP and morphogenesis. In the "PCP-first scenario" (top), the self-organising capacity of core PCP protein interactions creates tissue polarity without any specific cue to specify the direction. The advantage of coordinated polarity of cilia movement for filter-feeding, swimming or morphology could have favoured directed symmetry breaking using reliable cues. These could have initially been provided by environmental or mechanical constraints before the recruitment of localised Wnt binding to Fz to assure PCP orientation, and then the emergence of downstream β-catenin signalling.

Conservation of the PCP orientation and morphogenesis mechanisms in metazoans

In the *Drosophila* wing disc, local membrane-tethered Wg/Wnt4 in the wing margin can globally orient the PCP (Wu et al., 2013; Yu et al., 2020). In vertebrate tissues, Wnt ligands either act as global cues for PCP orientation (Chu and Sokol, 2016; Gros et al., 2009; Minegishi et al., 2017), or triggers involved in the body elongation (Heisenberg et al., 2000; Kilian et al., 2003; Tada and Smith, 2000; Wallingford and Harland, 2001). The role of Wnt for PCP orientation is thus common across cnidarians and bilaterians. The Wnt subtypes employed in PCP orientation are, however, not conserved. The Wnt gene family gained its complexity prior to the bilaterian-cnidarian bifurcation (Kusserow et al., 2005). Then, different Wnt subtypes were likely co-opted in different taxa and developmental contexts. In zebrafish, PCP orientation by wnt11 and core PCP regulation are distinctive in mechanosensory hair cell orientation (Acedo et al., 2019). The modifiable two-step mechanism of Wnt-driven PCP orientation is likely common in metazoans.

Other cues than Wnt also function as PCP orientation cues in the animal kingdom. In the developing *Drosophila* eye and abdomen, another signalling module comprising the atypical cadherins Fat(Ft) and Dachshous(Ds) and kinase Four-jointed(Fj) regulate the localisation of core PCP complexes via gradients of Ds and Fj (Strutt and Strutt, 2021; Yang et al., 2002). Cnidarians have conserved Ft and Ds ortholog genes. In hydra Fat-like protein (HyFat1) is implicated in epithelial cell alignment and adhesion (Brooun et al., 2020). In *Clytia*, inhibition of Fat-like (CheFat1) and Ds (CheDs) disrupted global PCP coordination, and the Ds disruption was restored by Wnt3 expression (Fig.S3), suggesting that these molecules participate in global PCP orientation in a manner yet to be studied. Besides molecular signalling cues, mechanical constraints or fluid flow are also known to direct PCP (Aigouy et al., 2010; Chien et al., 2022; Guirao et al., 2010; Mitchell et al., 2007; Olguin et al., 2011) and can potentially act as the PCP symmetry-breaking factor. The availability of alternative PCP orientation cues might have allowed evolutionary flexibility to link PCP to available structures, such as the cell polarity scaffold of eggs. While tissue polarity orientation by PCP is strongly conserved, the cellular mechanisms underlying the morphogenetic movement are highly variable. Oriented cell division or convergent extension are likely major driving forces of blastula elongation prior to gastrulation in *Podocoryne carnea* (Momose and Schmid, 2006), another hydrozoan jellyfish. In contrast, in *Clytia*, elongation likely involves PCP-dependent intercalation of both ectoderm (Byrum, 2001) and ingressing endoderm (Sande et al., 2020) cells during gastrulation. In *Nematostella*, hydraulic pressure created by body wall muscle, organised along the body axis, drives the elongation (Stokkermans et al., 2022) (Fig. S4)

Body axis symmetry-breaking in the evolutionary history of metazoa

The dual role of Wnt3 and the two-step PCP orientation mechanism revealed here are particularly intriguing in the context of understanding the emergence of animal body axes during evolution. The metazoan ancestor is considered to have taken a multicellular state such as a “choanoblastaea” from a unicellular ancestor (Arendt et al., 2015; Nielsen, 2008) (Fig. 6B). Then metazoans acquired a body axis and germ layers and formed a hypothetical ancestral state such as Haeckel’s “gastraea” (Haeckel, 1873) or Hyman’s “planuloid-acoeloid” (Hyman, 1951). Early multicellular ancestors thus acquired morphological axis patterning and coupled it with regional gene expression. It is tempting to speculate that the β -catenin/TCF-dependent local gene expression and morphogenetic patterning coordinated by core PCP proteins were independently acquired and coupled by co-opting Wnt ligands and signal transducers from one pathway to the other. For example, Wnt, Fz and Dsh might have first been employed in the Wnt/ β -catenin

pathway, then recruited to a primordial PCP regulatory system (Fig. 6B). Genome analyses indicate that PCP effectors and a part of core PCP components such as Cdc42, RhoA, ROCK, Prickle or Inversin are evolutionarily older than metazoans (Lapébie et al., 2011) (Fig. S5, Table S1). Wnt-Fz-Dsh would then have been recruited to create morphogenesis machinery directed by PCP based on the existing local gene expression by Wnt/ β -catenin signalling. Alternatively, core PCP interactions and Wnt-driven PCP orientation may be more ancestral with Wnt-Fz-Dsh used in PCP orientation prior to its involvement in “canonical” Wnt signalling, acquired later through recruitment of the cell adhesion molecule β -catenin and its interaction with the DNA-binding TCF. Comparisons of the genomic repertoires for these components between species from across the metazoan phylogeny do not allow us to conclude which modules were acquired first (Fig.S5, S6, Table S1). In the outgroups of the cnidaria-bilateria clade, a full set of core PCP proteins are present in a sponge *Oscarella carmela* (Lapébie et al., 2011; Nichols et al., 2006; Schenkelaars et al., 2016), and full Wnt/ β -catenin pathways were identified in both Porifera and Ctenophora (Holzem et al., 2024), while multiple cases of loss of Stbm/Fmi occurred in Porifera and potentially in Ctenophora (Fig.S5, Table S1).

PCP is a fundamental character in multicellular tissue organisation seen wide across metazoans (Fig. S5, S6) and may have been particularly important in the metazoan ancestor with lower anatomical complexity. It is known that a rotational polarity of the flagellar basal body is coordinated across the spheroid of green alga *Volvox carteri*, which thus independently acquired PCP-like character (Hoops, 1993) (Fig.S6), suggesting that a PCP equivalent evolved independently in *Volvox*. PCP-driven axis evolution is an attractive scenario and can explain how an initial “choanoblastaea” multicellular ancestor acquired coordinated cilia/flagella flow. For example, PCP regulation might initially have improved filter-feeding efficiency in a choanoblastaea ancestor (Cavalier-Smith, 2017; Nielsen, 2008). Little notice has been paid to the implication of the PCP pathway and cellular basis of tissue patterning in metazoan body plan evolution compared to the Wnt/ β -catenin pathway. A comparative cell biology, or “Cellular Evo-Devo” approach, of the PCP/tissue polarities in diverse metazoans, in particular “basal metazoans” including Cnidaria, Ctenophora and Porifera, will be the key to addressing how symmetry-breaking to create a body axis arose during evolution.

Materials and Methods

Key resources table

Reagent type or resource	Designation	Source or reference	Identifiers	Additional information
Morpholino oligo	Wnt3MO	Momose et al. 2008	EU374721.1	CCAAAACACACCAGTGTGCGAGCCAT
Morpholino oligo	Fmi-MO	This work	JQ439002.1	CCTCAAGCCATCTGAGCTTCATTTT
Morpholino oligo	Stbm-MO	Momose et al, 2012	JQ439008.1	TCACTCCATCATCAAAATCATCCAT
Morpholino oligo	Fat-like-MO	This work	JQ439001.1	CATCAAAATGTGAGACTTACCAGCC

Morpholino oligo	Ds-MO	This work	JQ438999.1	TACGGTGATGGATAGTTCATCTTTC
Dye	Phalloidin Alexa Fluor™ 488	Invitrogen	A12379	4 unit/ml 0.13 µM in PBS
Dye	Phalloidin Alexa Fluor™ 647	Invitrogen	A22287	4 unit/ml 0.13 µM in PBS
antibody	anti-Par3	Merck Millipore	07-330	
antibody	anti-γ Tubulin	SIGMA	T5326 (GTU-88)	2 µg/ml (x1/500 dilution)
antibody	anti-centrin	Merck Millipore	04-1624 (20H5)	
mRNA	Wnt3	Momose et al. 2008	XLOC_001931	Wildtype (3 bp mismatch in MO target) 100 ng/µl
mRNA (mutant)	dnTCF	This work	XLOC_007658	Dominant negative N-terminal 27 a.a. deletion 400 ng/µl
mRNA (mutant)	CA-β-cat	This work	XLOC_003560	Constitutively active S97A, T101A, S105A 20 ng/µl
mRNA (mutant)	RhoABC-DN	This work	XLOC_044216	Dominant negative T19N 10 ng/µl
mRNA (mutant)	RhoABC-CA1	This work	XLOC_044216	Constitutively active G14V 10 ng/µl
mRNA (mutant)	RhoABC-CA2	This work	XLOC_044216	Constitutively active Q63L 10 ng/µl
Fluorescence protein	PH-Venus	This work	U09117.1	mCherry fusion protein 100 ng/µl
Fluorescence protein	Poc1-Venus	This work	HM010924.1	Venus fusion protein 2 ng/µl
Animal	<i>Clytia hemisphaerica</i>	EMBRC-Fr	Z4C2	Male strain discontinued
Animal	<i>Clytia hemisphaerica</i>	EMBRC-Fr	Z4B	Female strain discontinued
Animal	<i>Clytia hemisphaerica</i>	EMBRC-Fr	Z23	Male (Z4C2 x Z4B)
Animal	<i>Clytia hemisphaerica</i>	EMBRC-Fr	Z30	Female (Z4C2 x Z4B)

Animal culture and embryo preparation

Animal culture methods are described in (Lechable et al., 2020 [↗](#)). *Clytia hemisphaerica* Laboratory strains were provided by the Service Aquariologie of CRB (IMEV-FR3761, Sorbonne Université/CNRS contact: Axel Duchene axel.duchene@imev-mer.fr) as an EMBRC-Fr service (<https://www.embrc-france.fr/↗>).

Daily spawning was induced by light with a 24-hour day-night cycle. Eggs and sperm were separately collected from jellyfish and transferred into 10 cm glass dishes two hours after the light stimulation. Eggs were fertilised less than one hour after the spawning. To acquire EG stages, embryos were incubated at 16°C for 18 hours, which corresponds to the 12~13 hpf stage at the standard culture temperature (18°C).

Microinjection

Morpholinos and mRNA were microinjected into unfertilised eggs or blastomeres at 4-cell ~ 16-cell stages using an Eppendorf Femtojet microinjector with microcapillaries pulled with Narishige PN-30 needle puller. The tip of the needle was broken by touching it to a glass surface scratched by a diamond pen to open the needle tip to give an outer diameter of 2 to 4 µm. Eggs were aligned in a microinjection chamber made by casting and polymerising PDMS in a negative mould printed with a Form3 3D printer. 2~3% of the egg/cell volume of solution was injected by inserting a microcapillary needle with constant back pressure (Femtojet compensation pressure). It typically takes 0.5~1 seconds for each egg at 5~10 kPa when the needle has an appropriate opening size.

The volume of injection for each injection was visually controlled under an Olympus SZX6 stereomicroscope with SZ-ILLT oblique illumination. 3~5 mg/ml lysine fixable 10 kDa Dextran labelled with Alexa647 or Cascade Blue were co-injected with Wnt3 mRNA and Stbm-MO/Fmi-MO for the lineage tracking in the Wnt3-Rescue experiment and PCP mosaic experiments.

Plasmid construction

Mutant and fluorescent protein templates were constructed in pCX3 vector, which contains short 5' and 3'UTR sequences from CheStbm mRNA. Wildtype CheWnt3 with 3 mispaired nucleotide residues at the Morpholino target and CheTCF lacking amino acid residues 2-27 (dnTCF) as a dominant negative form (lacking β -catenin binding site) of human TCF-4 (Korinek et al., 1997 [\[1\]](#)) were amplified by PCR and cloned into the vector. A constitutively active form of Che- β -catenin (CA- β -cat) was created by introducing S97A, T101A and S105A mutations corresponding to the constitutively active mice β -catenin mutant lacking a GSK3- β phosphorylation target (Baba et al., 2006 [\[2\]](#)). Constitutively active forms (CA1: G14V, CA2: Q63L) that gave identical phenotypes and a dominant form (DN: T19N) of CheRhoABC were designed based on the identical vertebrate RhoA mutants with the known phenotypes. These point mutations were introduced into wildtype cDNA clones in pCX3 vector using the QuickChange site-directed mutagenesis kit (Stratagene).

Poc1-mCherry and PH-domain-Venus fusions were constructed in a transgenic pMiniTol2-ACT2 vector (Weissbourd et al., 2021 [\[3\]](#)) using NEBuilder Assembly kit from individually amplified PCR products or synthetic genes. mRNA was synthesised in vitro using mMessage mMachine T3 kit and poly(A)-tailing kit (Thermo Fisher) from PCR products amplified from the plasmids.

Immunostaining

Immunofluorescent visualisation of PCP by gamma-tubulin antibody and phalloidin staining was described previously (Momose et al., 2012 [\[4\]](#)). List of the antibodies are in the Supplementary method. In brief, the embryos were fixed with 4% Paraformaldehyde in 0.1 M Hepes, 50 mM EGTA, 80 mM Maltose, 10 mM MgSO₄, 0.1% Triton X-100 and washed in PBST (1x PBS with 0.1% Triton X-100) then incubated in antibody solution in 1% BSA fraction V in PBST, followed by three times of washes for at least 5 minutes in PBST. After removing the detergent by washing in PBS, actin was stained with fluorescently labelled phalloidin in PBS (0.13 μ M). After staining, embryos were cleared and mounted in Citiflour AF-1 diluted with an equal volume of PBS and scanned using Leica SP5 or Stellaris confocal microscopes with 63x objectives. The surface of the ciliated epidermis was Z-scanned (up to 10 μ m) in the entire 246 x 246 μ m field of view and z-stacked to visualise cell contour with apical cortical actin and basal body, typically at 0.081 μ m resolution in XY-axis and 0.5~1 μ m resolution in Z-axis.

Image analysis

The x-y coordinates of the apical surface centroid (x_c, y_c) of each cell and basal body of its cilium (x_b, y_b) are identified from the z-stacked confocal images. Detailed procedures and ImageJ scripts are described in the supplementary method. The polarity angle θ for each cell was calculated as

$$\theta = \text{atan2}(y_c - y_b, x_c - x_b)$$

The mean orientation (arrow in the radar plot) and circular standard deviation (circ. s.d.) are calculated (Berens, 2009 [\[5\]](#)). A Matlab script was used to summarise the PCP orientation and statistics (Supplementary Method). The morphology of embryos in three dimensions (**Figure. 5D, H** [\[6\]](#)) is reconstructed from a series of cortical actin confocal images by building a surface of the volume containing the cortical actin signals using Imaris software (Oxford Instruments).

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Additional information

Abbreviations

- PCP: planar cell polarity
- EG: early gastrula
- LG: late gastrula
- MO: Morpholino
- OA: Oral-Aboral

Additional files

Supplementary Method [↗](#)

Supplementary Movie [↗](#)

Supplementary Figures 1-6 [↗](#)

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Reviewer #1 (Public review):

Summary:

This noteworthy paper examines the role of planar cell polarity and Wnt signalling in body axis formation of the hydrozoan *Clytia*. In contrast to the freshwater polyp *Hydra* or the sea anemone *Nematostella*, *Clytia* represents a cnidarian model system with a complete life cycle (planula larva-polyp-medusa). In this species, classical experiments have demonstrated that a global polarity is established from the oral end of the embryos (Freeman, 1981). Prior research has demonstrated that Wnt3 plays a role in the formation of the oral organiser in *Clytia* and other cnidarians, acting in an autocatalytic feedback-loop with β -catenin. However, the question of whether and to what extent an oral-aboral gradient of Wnt activity is established remained unanswered. This gradient is thought to control both tissue differentiation and tissue polarity. The planar cell polarity (PCP) pathway has been linked to this polarity, although it is generally considered to be β -catenin independent.

Comments on major strengths and weaknesses:

Beautiful and solid experiments to clarify the role of canonical Wnt signalling and PCP core factors in coordinating planar cell polarity of *Clytia*. The authors have conducted a series of sophisticated experiments utilising morpholinos, mRNA microinjections and immunofluorescent visualisation of PCP. The objective of these experiments was to address the function of Wnt3, β -catenin and PCP core proteins in the coordination of the global polarity of *Clytia* embryos. The authors conclude that PCP plays a role in regulating polarity along the oral-aboral axis of embryos and larvae. This offers a conceivable explanation for how polarity information is established and distributed globally during *Clytia* embryogenesis, with implications for our understanding of axis formation in cnidarians and the evolution of Wnt signalling in general. - While the experiments are well-designed and executed, there are some criticisms, questions or suggestions that should be addressed.

(i) Wnt3 cue and global PCP. PCP has been described in detail in a previous paper on *Clytia* (Momose et al, 2012): its orientation along the oral-aboral body axis (ciliary basal body

positioning studies), and its function in directional polarity during gastrulation (Stbm-, Fz1-, and Dsh-MO experiments). I wonder if this part could be shortened. What is new, however, are the knockdown and Wnt3-mRNA rescue experiments, which provide a deeper insight into the link between Wnt3 function in the blastopore organiser as a source or cue for axis formation. These experiments demonstrate that the Wnt3 knockdown induces defects equivalent to PCP factor knockdown, but can be rescued by Wnt3-mRNA injection, even at a distance of 200 μm away from the Wnt-positive area. The experimental set-up of these new molecular experiments follows in important aspects those of Freeman's experiments of 1981 (who in turn was motivated to re-examine Teissier's work of 1931/1933 ...). Freeman did not use the term "global polarity" but the concept of an axis-inducing source and a long-range tissue polarity can be traced back to both researchers.

(ii) PCP propagation and β -catenin. The central but unanswered question in this study focuses on the interaction between Wnt3 and PCP and the propagation of PCP. Wnt3 has been described in cnidarians but also in vertebrates and insects as a canonical Wnt interacting with β -catenin in an autocatalytic loop. The surprising result of this study is that the action of Wnt3 on PCP orientation is not inhibited in the presence of a dominant-negative form of CheTCF (dnTCF) ruling out a potential function of β -catenin in PCP. This was supported by studies with constitutively active β -catenin (CA- β -cat) mRNA which was unable to restore PCP coordination nor elongation of Wnt3-depleted embryos but did restore β -catenin-dependent gastrulation. Based on these data, the authors conclude that Wnt3 has two independent roles: Wnt/ β -catenin activation and initial PCP orientation (two step model for PCP formation). However, the molecular basis for the interaction of Wnt3 with the PCP machinery and how the specificity of Wnt3 for both pathways is regulated at the level of Wnt-receiving cells (Fz-Dsh) remains unresolved. - Also, with respect to PCP propagation, there is no answer with respect to the underlying mechanisms. The authors found that PCP components are expressed in the mid-blastula stage, but without any further indication of how the signal might be propagated, e.g., by a wavefront of local cell alignment. Here, it is necessary to address the underlying possible cellular interactions more explicitly.

(iii) The proposed two step model for PCP formation has important evolutionary implications in that it excludes the current alternate model according to which a long-range Wnt3-gradient orients PCP ("Wnt/ β -catenin-first"). Nevertheless, the initial PCP orientation by Wnt3 - as proposed in the two-step-model - is not explained at all on the molecular level. Another possible, but less well discussed and studied option for linking Wnt3 with PCP action could be a role of other Wnt pathways. The authors present compelling evidence that Wnt3 is the most highly expressed Wnt in *Clytia* at all stages of development. The authors convincingly show that Wnt3 is the most highly expressed Wnt in *Clytia* at all stages of development (Fig. S1). However, Wnt7 is also more highly expressed, which makes it a candidate for signal transduction from canonical Wnts to PCP Wnts. An involvement of Wnt7 in PCP regulation has been described in vertebrates (<http://dx.doi.org/10.1016/j.celrep.2013.12.026>). This would challenge the entire discussion and speculation on the evolutionary implications according to which PCP Wnt signaling comes first (PCP-first scenario") and canonical Wnt signaling later in metazoan evolution.

(iv) The discussion, including Figure 6, is strongly biased towards the traditional evolutionary scenario postulating a choanozoan-sponge ancestry of metazoans. Chromosome-linkage data of pre-metazoans and metazoans (Schulz et al., 2023; <https://doi.org/10.1038/s41586-023-05936-6>) now indicate a radically different scenario according to which ctenophores represent the ancestral form and are sister to sponges, cnidarians and bilaterians (the Ctenophora-sister hypothesis). This also has implications for the evolution of Wnt signalling, as discussed in the recent Nature Genetics Review by Holzem et al. (2024) (<https://doi.org/10.1038/s41576-024-00699-w>). Furthermore, it calls into question the hypothesis of a filter-feeding multicellular gastrula-like ancestor as proposed by Haeckel (Maegle et al., 2023). These papers have not yet been referenced, but they would provide a more robust discussion.

General appraisal:

The authors have carefully addressed all important points raised in this review. Aims and results support their conclusions.

Impact of the work, utility of methods and data:

As stated above, there will be a major impact on our understanding of the role of Wnt signaling in gradient formation and particularly the role of non canonical wnt signaling. As mentioned above, this will have a major impact on our understanding of the role of Wnt signalling in gradient formation, particularly the role of non-canonical Wnt signalling. - It will also be important to better understand the role of Wnt-Frizzled interactions in these basal organisms, as cnidarians have a smaller repertoire of Frizzled receptors compared to the relatively complete repertoire of Wnt subfamilies. This may imply that Wnt 3 is active in both canonical and PCP.

Additional context:

With regard to the question of the evolution of the body plan and Wnt signalling, it would be helpful and important for readers unfamiliar with cnidarians to know that the Hydrozoa/Medusozoa, to which *Clytia* belongs, are an "evolutionary derived group" within the Cnidaria, as opposed to the Anthozoa (e.g. sea anemone *Nematostella*). Hydrozoans possess planula larvae that are devoid of a mouth and any form of feeding mechanism, relying instead on the yolk of a fertilised egg for sustenance. The substantial divergence between the Anthozoa and Medusozoa was accompanied by significant gene reductions within the Medusozoa, which likely exerts an influence on the evolution of Wnt signalling in this group as well. This should not detract from the value of the work, but may help to put it in perspective.

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Reviewer #2 (Public review):

Summary:

Canonical Wnt signaling has previously been shown to be responsible for correct patterning of the oral-aboral axis as well as germ layer formation in several cnidarians. The post-gastrula stage, the planula larvae is not only elongated, it has a specific swimming direction due to the decentralized cellular positioning and slanted anchoring of the cilia. This, in turn, is in most other animals the result of a Wnt-Planar-cell polarity pathway. This paper by Uveira et al investigates the role of Wnt3 signaling in serving as a local cue for the PCP pathway which then is responsible for the orientation of the cilia and elongation of the planula larva of the hydrozoan *Clytia hemisphaerica*. Wnt3 was shown before to activate the canonical pathway via β -catenin and to act as an axial organizer. The authors provide compelling evidence for this somewhat unusual direct link between the pathways through the same signaling molecule, Wnt3. In conclusion, they propose a two-step model: 1) local orientation by Wnt3 secretion 2) global propagation by the PCP pathway over the whole embryo.

Strengths:

In a series of elegant and also seemingly sophisticated experiments, they show that Wnt3 activates the PCP pathway directly, as it happens in the absence of canonical Wnt signaling (e.g. through co-expression of dnTCF). Conversely, constitutive active β -catenin was not able to rescue PCP coordination upon Wnt3 depletion, yet restored gastrulation. This uncouples

the effect of Wnt3 on axis specification and morphogenetic movements from the elongation via PCP. Through transplantation of single blastomeres providing a local source of Wnt3, they also demonstrate the reorganization of cellular polarity immediately adjacent to the Wnt3 expressing cell patch. These transplantation experiments also uncover that mechanical cues can also trigger the polarization, suggesting a mechanotransduction or direct influence on subcellular structures, e.g. actin fiber orientation.

This is a beautiful and elegant study addressing an important question. The results have significant implications also for our understanding of the evolutionary origin of axis formation and the link of these two ancient pathways, which in most animals are controlled by distinct Wnt ligands and Frizzled receptors. The quality of the data is stunning and the paper is written in a clear and succinct manner. This paper has the potential to become a widely cited milestone paper.

Weaknesses:

I can not detect any major weaknesses. The work only raises a few more follow-up questions, which the authors are invited to comment on.

I acknowledge the revisions made by the authors. Some open questions remain that need to be addressed in future work, and I accept the limitations of this study, as argued by the authors. Besides the elegant and high-quality experiments, I also appreciate the thoughtful and inspiring discussion.

<https://doi.org/10.7554/eLife.104508.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):

(1) Wnt3 cue and global PCP. PCP has been described in detail in a previous paper on Clytia (Momose et al, 2012): its orientation along the oral-aboral body axis (ciliary basal body positioning studies), and its function in directional polarity during gastrulation (Stbm-, Fz1-, and Dsh-MO experiments). I wonder if this part could be shortened. What is new, however, are the knockdown and Wnt3-mRNA rescue experiments, which provide a deeper insight into the link between Wnt3 function in the blastopore organiser as a source or cue for axis formation. These experiments demonstrate that the Wnt3 knockdown induces defects equivalent to PCP factor knockdown, but can be rescued by Wnt3-mRNA injection, even at a distance of 200 μ m away from the Wnt-positive area. The experimental set-up of these new molecular experiments follows in important aspects those of Freeman's experiments of 1981 (who in turn was motivated to re-examine Teissier's work of 1931/1933 ...). Freeman did not use the term "global polarity" but the concept of an axis-inducing source and a long-range tissue polarity can be traced back to both researchers.

We appreciate the reviewer's insightful comments for evolutionary biology and cnidarian developmental biology.

Concerning the presentation of the basic PCP structure of *Clytia* embryo epidermal cells, we prefer to retain this section unless there is a strict limit on manuscript length. These experiments provide background information necessary to establish the biological system for the readers. The structures of cells, notably cell adhesion, cilia, and the cytoskeleton, are essential components of this system.

We have restored sentences concerning the historical contributions of Freeman and Teissier from a previous version of the manuscript.

Freeman's work offered two key insights. The first is the concept that cell polarity spreads and self-organizes over the distances revealed by the tissue orientation of aggregate embryonic cells (Freeman, 1981 <https://doi.org/10.1007/BF00867804>), which was termed "global polarity" in a review by Primus and Freeman (2004 <https://doi.org/10.1002/bies.20031>). This concept closely resembles the modern understanding of PCP coordination mechanisms mediated by core PCP interactions. Remarkably, Freeman proposed this idea in the early 1980s, at the same time of the first characterization of PCP mutants in *Drosophila* (Gubb and Garcia-Bellido 1982). The second is the role of egg polarity in defining the axis. Freeman demonstrated that the position of the first cleavage furrow predicts the oral-aboral axis by a series of sophisticated experiments. This was a milestone for the studies of cnidarian body axis development.

However, some of Freeman's interpretations were misleading. In the 1981 paper, he stated:

"Polarity

Other work that I have done has established that the anterior-posterior axis of the planula is set up at the time of the first cleavage; the site where cleavage is initiated specifies the posterior pole of this axis (Freeman 1980). The experiment reported here in which embryos were cut into halves and each half regulated to form a normal planula with the same polarity properties as the embryo it is from provides evidence that these polarity properties are remarkably stable at all developmental stages tested ranging from 4 cell to postgastrula embryos."

Freeman hypothesised that cell polarity at the 2- or 4-cell stage, referred to as the "polarity of first cell cleavage," is directly inherited as the global polarity observed in later developmental stages.

In the review by Primus and Freeman (2004), two hypotheses were introduced: (1) maternally localised factors, such as mRNA, determine the axis, and (2) cell polarity of cleavage furrow formation, is inherited to later stages and determines the axis. Freeman described these two hypotheses as mutually exclusive. However, we now know that cell polarity at early cleavage stages does not directly contribute to global polarity/PCP. Instead, Wnt/ β -catenin signaling is regionally activated by maternally localised mRNAs distributed along the egg polarity (Momose, 2007; Momose, 2008), which maintain Wnt3 localisation and direct morphological axis patterning. Our study shown in this article unified these hypotheses.

On the second point, as the reviewer noted, Freeman indeed revisited the work of Georges Teissier (Teissier, 1931), who conducted similar experiments on *Amphibetia* embryos. It was Teissier who first described how the egg polarity is preserved in later stages and defines the axis. Teissier, however, carefully avoided asserting continuity between egg and blastula polarities, allowing for the possibility of "rétablissement" (re-establishment). As Teissier stated:

"...On constate, en second lieu, que la polarité de l'œuf se conserve dans chacun de se fragment et que le maintien ou le rétablissement de cette polarité sont indispensables à un développement normal. Un fragment d'œuf ou de morula n'a aucune partie ni aucun blastomère qui soit rigoureusement déterminé comme endoderme, mais possède, par contre, un pôle antérieur et un pôle postérieur bien définis....

Mais cette proposition, qui ne semble pourtant guère dépasser la simple constatation des faits, soulève de grave difficulté. Elle donne en effet à la polarité, propriété encore bien

mystérieuse, un rôle morphogénétique de premier ordre et implique des conséquences trop importantes pour qu'on puisse l'accepter sans un très sérieux examen.

Comme je ne pense pas que les questions relatives à la nature des localisation germinales, à l'existence et au fonctionnement des organisateurs de l'œuf des Coelentérés, puissant, dans l'état actuel de nos connaissances, être discutées utilement, je ne veux voir dans la proposition précédente qu'une façon commode et tout provisoire de systématiser les faits."

English translation:

"We note also that the polarity of the egg is preserved in each fragment and that the maintenance or re-establishment of this polarity is essential for normal development. A fragment of egg or morula has no part or blastomere that is rigorously determined as endoderm, but has, on the other hand, a well-defined anterior and posterior pole....

But this proposition, which hardly seems to go beyond the simple observation of facts, raises serious difficulties. It gives polarity, still a mysterious property, a morphogenetic role of the first order, and implies consequences too important to be accepted without very serious examination.

As I do not believe that questions concerning the nature of germinal localisation, or the existence and functioning of the egg organisers in Coelenterates, can, in the present state of our knowledge, be usefully discussed, I prefer only to see in the foregoing proposition a convenient and very provisional way of systematising the facts."

Teissier, G. (1931). Étude Expérimentale du Développement de Quelques Hydriaires. Ann. Sc. Nat. Zool 14, 5–59.

Teissier's interpretation and caution were reasonable.

Our work connects recent molecular research on axis specification mechanisms in cnidarians with the classic experimental studies of Freeman and Teissier. We believe it is essential to present and acknowledge their conceptual contributions. We have updated the Discussion to include these points.

(2) PCP propagation and β -catenin. The central but unanswered question in this study focuses on the interaction between Wnt3 and PCP and the propagation of PCP. Wnt3 has been described in cnidarians but also in vertebrates and insects as a canonical Wnt interacting with β -catenin in an autocatalytic loop. The surprising result of this study is that the action of Wnt3 on PCP orientation is not inhibited in the presence of a dominant-negative form of CheTCF (dnTCF) ruling out a potential function of β -catenin in PCP. This was supported by studies with constitutively active β -catenin (CA- β -cat) mRNA which was unable to restore PCP coordination nor elongation of Wnt3-depleted embryos but did restore β -catenin-dependent gastrulation. Based on these data, the authors conclude that Wnt3 has two independent roles: Wnt/ β -catenin activation and initial PCP orientation (two-step model for PCP formation). However, the molecular basis for the interaction of Wnt3 with the PCP machinery and how the specificity of Wnt3 for both pathways is regulated at the level of Wnt-receiving cells (Fz-Dsh) remain unresolved. Also, with respect to PCP propagation, there is no answer with respect to the underlying mechanisms. The authors found that PCP components are expressed in the mid-blastula stage, but without any further indication of how the signal might be propagated, e.g., by a wavefront of local cell alignment. Here, it is necessary to address the underlying possible cellular interactions more explicitly.

The question of how Wnt3 interacts with the core PCP complex remains open for future investigation. An obvious hypothesis is that one of the Frizzled receptors binds Wnt3 ligands. For additional details, please refer to the response to Reviewer 2's comment. Regarding other

non-classic Wnt receptors, studies in the developing mouse limb have demonstrated that a Wnt5a gradient controls PCP polarisation via ROR receptors and graded Strabismus phosphorylation (Gao et al., 2011, <https://doi.org/10.1016/j.devcel.2011.01.001>). However, in this context, the Wnt5a gradient influences the frequency of polarised cells rather than PCP orientation. In *Clytia*, we performed gene knockdown experiments targeting ROR and RYK receptors using Morpholinos but did not observe any effect on axial patterning, suggesting that these receptors are unlikely to be involved in Wnt3 interaction.

Concerning PCP propagation mechanisms, these are well-characterized in both *Drosophila* and vertebrates and conserved across taxa. The localised Fz-Fmi complex at the apical cortex of a cell interacts with the oppositely localised Stbm-Fmi complex in neighbouring cells, enabling coordination of PCP between directly adjacent cells. This interaction provides a comprehensive explanation for PCP propagation mechanisms. In *Drosophila*, the “domineering non-autonomy” effect is a well-documented phenomenon where PCP orientation autonomously propagates from core PCP mutant mosaic patches. Overall, PCP propagation is a conserved and robust mechanism across metazoans.

*(3) The proposed two-step model for PCP formation has important evolutionary implications in that it excludes the current alternate model according to which a long-range Wnt3-gradient orients PCP ("Wnt/ β -catenin-first"). Nevertheless, the initial PCP orientation by Wnt3 - as proposed in the two-step model - is not explained at all on the molecular level. Another possible, but less well-discussed and studied option for linking Wnt3 with PCP action could be the role of other Wnt pathways. The authors present compelling evidence that Wnt3 is the most highly expressed Wnt in *Clytia* at all stages of development. The authors convincingly show that Wnt3 is the most highly expressed Wnt in *Clytia* at all stages of development (Figure S1). However, Wnt7 is also more highly expressed, which makes it a candidate for signal transduction from canonical Wnts to PCP Wnts. An involvement of Wnt7 in PCP regulation has been described in vertebrates (<http://dx.doi.org/10.1016/j.celrep.2013.12.026>). This would challenge the entire discussion and speculation on the evolutionary implications according to which PCP Wnt signaling comes first (PCP-first scenario) and canonical Wnt signaling later in metazoan evolution.*

First of all, we apologise that the expression profile of Wnt7 originally provided in Figure S1 was incorrect; Wnt7 is not expressed in the embryonic stage. The error came from the accession number XLOC_034538 assigned to two transcripts, Wnt7 and Ataxin10, in the published genome assembly. Once the expression profile is revised in this light, the data are consistent with the in situ hybridisation data published in Momose et al. (2012, <https://doi.org/10.1242/dev.084251>). Wnt3 is the only Wnt ligand detectable between egg and gastrula stages. We appreciate the reviewer highlighting this issue and have corrected Figure S1

If we understand correctly, the reviewer raises the possibility that Wnt3's downstream canonical Wnt/ β -catenin pathway activates the expression of other Wnt genes, which in turn orient the PCP. Indeed, we showed that the expression of Wnt1 (previously called WntX2), Wnt2 (WntX1A), Wnt5 and Wnt6 (Wnt9) all becomes undetectable at the planula stage following Wnt3-MO injection (Momose et al., 2012). So, it is a reasonable concern.

This possibility can be excluded because the canonical pathway activation by CA- β -cat does not restore PCP in Wnt3-MO-injected embryos and Wnt3 can orient PCP without Wnt/ β -catenin pathway activity in the presence of dominant negative TCF (dnTCF). Concerning Wnt1b and Wnt11b, these transcripts are maternally stored and even more abundant than Wnt3. However, we can conclude that these do not have any role in axis patterning based on the complete axis loss in Wnt3-MO morphants.

Lastly, it should of course be remembered that the chronological order of characters appearing in a developmental process does not necessarily reflect their appearance in evolution from ancestral to modern.

(4) The discussion, including Figure 6, is strongly biased towards the traditional evolutionary scenario postulating a choanozoan-sponge ancestry of metazoans. Chromosome-linkage data of pre-metazoans and metazoans (Schulz et al., 2023; <https://doi.org/10.1038/s41586-023-05936-6>) now indicate a radically different scenario according to which ctenophores represent the ancestral form and are sister to sponges, cnidarians and bilaterians (the Ctenophora-sister hypothesis). This has also implications for the evolution of Wnt signalling, as discussed in the recent Nature Genetics Review by Holzem et al. (2024) (<https://doi.org/10.1038/s41576-024-00699-w>). Furthermore, it calls into question the hypothesis of a filter-feeding multicellular gastrula-like ancestor as proposed by Haeckel (Maegle et al., 2023). These papers have not yet been referenced, but they would provide a more robust discussion.

I overlooked the excellent work of Holzem and colleagues. I appreciate this suggestion. The work, unfortunately, focusses mainly on the Wnt/ β -catenin pathway. The PCP pathway consists of not only core PCP (Fmi Stbm, Pk, Dgo, Fz and Dsh) but many other components, such as Rho GTPase, which are all dealt with as "PCP" in this review. While the full set of core PCP is present only in the phylum Cnidaria and bilaterians, Pk and Dgo are present in choanoflagellate and Rho GTPase or ROCK are present even in Fungi (Lapébie et al, 2011 DOI 10.1002/bies.201100023). Holzem et al., described PCP as absent in ctenophores, likely based on the lack of Fmi/Stbm, while claiming its presence in fungi based on Rho GTPase and ROCK. This led to their argument that the Wnt/ β -catenin pathway is more ancestral, supported by the absence of PCP components in ctenophores alongside the ctenophore-sister hypothesis.

This likely reflects the limited attention given to PCP in the metazoan evolutionary biology community. Our work sheds light on the importance of PCP regulation in metazoan evolution. In the revised Discussion, we emphasise this point together with the importance of cell biology studies in basal metazoans and compare them based on functional studies.

The observation of *Aiptasia*'s predatory "gastrula-like" larvae is indeed fascinating. Understanding how early metazoan ancestors obtained nutrients is a key to uncovering the origins of metazoans. However, the relevance of this work to metazoan evolution remains unclear. Predatory nutrient uptake is common among cnidarians, and the findings of Maegle et al. could suggest that the predatory gastrula-like state is ancestral, with the symbiotic state being derived, within Cnidaria, but does not notably support it in metazoa. Also, it has to be clarified how predation is defined. Fundamentally, there is little distinction between filter-feeding and predatory feeding regarding heterotrophy; both feeding types require digestive machinery. If active feeding behaviour is the essence of predation, this would be better addressed as an evolutionary neurobiology or neuroscience question. Another mystery is what the metazoan ancestors took as food if they were predatory; there has to be a non-predatorial metazoan, as a food, already present before them. Overall, Maegle's work seems premature to be incorporated into the metazoan evo-devo discussion. In either case, the standard approach would involve comparative studies across taxa. It will be interesting to see follow-up works on comparative and functional genomics of predatory/digestive machinery within phylum cnidaria and across metazoan, including sponge and ctenophores.

Reviewer #2 (Recommendations for the authors):

We appreciate the reviewer's expertise and recommendations regarding Wnt and PCP signalling. It would be our great pleasure if our work is seen and referenced by the cell

biology community using model animals.

(1) According to the 2-step model, one would expect that there is a temporal gradient in the spreading of the PCP from oral to aboral. Is there any indication for this?

The best indication of a spatial and temporal gradient of PCP establishment observed so far is at the blastula stage (Fig.2B). PCP gradually becomes coordinated starting at 9 hpf, when PCP is slightly better organised close to the Wnt3-positive area (oral) compared to distal (aboral) areas. We did live imaging with tagged Poc1 to track the positions of centrioles in each cell (Fig. 2E), but this did not provide any further information about the spreading of the PCP. We hypothesise that there is a delay between PCP polarisation—established through the subcellular localisation of core PCP components—and its structural manifestation as ciliary positioning and orientation. This delay likely varies between cells, preventing the formation of a precise spatial PCP wave. We hope in the future to address this temporal aspect by live-imaging of core PCP proteins labelled with fluorescent proteins.

(2) PCP is likely to be an all-or-nothing effect, while axial patterning is dose-dependent. Is there a critical dose of Wnt3 level required to kick off the PCP pathway?

We agree that the PCP phenotype is all-or-nothing. Although we did not perform a quantitative test, we have not seen any intermediate phenotypes in Wnt3-rescue experiments. In our experimental condition (100 ng/μl mRNA), the Wnt3 mRNA injection into a blastomere consistently restores the body axis (via PCP) of Wnt3-MO injected embryos. No axis restoration was observed at 1 ng/μl. At 10 ng/μl, some embryos showed a restored elongated axis, while others showed no axis. The volume of injection is not precisely controllable and can easily vary two-fold, so we assume the limit is somewhere around 10 ng/μl. This contrasts with endoderm rescue via Wnt/β-catenin activation by GSK-β-inhibitors (alsterpaullone) or the constitutively active β-catenin (CA-β-cat), which occurs in a dose-dependent manner (ex. Supplementary Figure S2).

(3) The key question left unaddressed is whether Wnt3 signals through one or two different Frizzled receptors? Which Frizzled receptors are candidates for this? Could they be knocked down to see which pathway (or both) is affected?

How Wnt3 orientates the PCP system is an extremely interesting question that needs to be answered, and we plan to address this in the future. In *Clytia*, four Frizzled genes have been identified in the genome: *CheFz1* (vertebrate counterpart of Fz1, 2, 3, 6 and 7), *CheFz2* (Fz5 and 8), *CheFz3* (Fz9/10) and *CheFz4* (Fz4). Knockdown of *CheFz1*, hereby called Fz1, by Morpholino showed a PCP phenotype (Momose 2012, supplementary data). For a long time, we have suspected that the most likely candidate for PCP mediation is *CheFz1*. The Wnt3-rescue experiment in *CheFz1*-blocked background (similar experiment to Figure 3E, F) could potentially have answered this question. No PCP orientation would be expected even near the Wnt3-mRNA injected area if *CheFz1* was the Wnt3 receptor for PCP orientation. Unfortunately, no reliable PCP phenotype was observed in this experiment, so this experiment was not included in the manuscript. We initially thought this was due to incomplete suppression of *CheFz1* mRNA translation by the Morpholino when used at sub-toxic doses. But we now favour the alternative explanation that Fz1 does not mediate the Wnt3 signal responsible for initiating PCP orientation. We have previously shown that Fz1 is required for the Wnt/β-catenin pathway (indicated by nuclear β-catenin localisation Momose 2007), which is then required to maintain Wnt3 expression. We cannot rule out that the PCP phenotype obtained previously following Fz1 knockdown (supplementary data in Momose 2012) is an indirect effect of Wnt3 downregulation.

In future work, we plan to test the PCP involvement of the other *Clytia* Frizzleds, notably CheFz2 and CheFz4, which are not present as maternal mRNAs but are zygotically expressed in the early gastrula stage. CheFz3 is unlikely to be a candidate because it is aborally localised and acts as a negative receptor for the Wnt/ β -catenin pathway (Momose 2007). Lastly, in unpublished experiments, no axial phenotype was obtained with ROR and RYK knockdown by Morpholino (T. Momose unpublished).

Based on these considerations, our current working hypothesis is that Wnt3 somehow stabilises or activates one of the Frizzled receptors acting as a core PCP protein in a polarised manner, likely at the oral side of each cell (Stbm is localised at the aboral side), which breaks the PCP symmetry and is propagated across the body axis.

A few lines have been added to the discussion regarding this point.

(4) Is there also PCP within the Wnt3 expressing domain? In other words, (and linked to question 2), does PCP require a certain concentration of Wnt3 or a gradient of Wnt3 in order to provide an orientation?

In the context of a simple Wnt3-MO rescue experiment, PCP is coordinated within the Wnt3-positive area. But this could be because PCP can propagate in both orientations, so it does not answer the question. In the Wnt3-rescue experiments in Fmi-MO and Stbm-MO embryos, PCP seemed better oriented close to the boundary between Wnt3-positive and -negative areas, in particular outside the Wnt3-positive area and rather uncoordinated deep in the middle of Wnt3-RNA positive area.

If Wnt3 expression is uniform across an embryo, as achieved by Wnt3-mRNA injection into the egg, the axis will be lost entirely (Momose 2008). We interpret these observations as indicating that Wnt3 expression "contrasts" (or steep gradients) act as the PCP orientation cue rather than a permissive manner.

In normal development, mRNA expression detected by in situ hybridisation has a slight gradient, but we do not have any information about the endogenous protein distribution.

We greatly appreciate the reviewer's insightful comments. A few sentences addressing points (2) and (4) have been added. The graphical models in Figures 4 and 6A have been updated. While these are relatively minor changes to the manuscript, they significantly impact future perspectives.

Minor comments:

(1) Labeling in some of the figures is too small and not legible, e.g. Figures 4E-H. Please check and improve.

Agreed. Some labelling was way too small (2.5 points). This has been corrected. The minimum font size is now 6-point for most labelling in the revised Figures.

(2) Page 13: ...and allow us to novel scenarios for PCP-driven axis symmetry breaking... seems to lack the verb "propose"

Corrected.

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